

The 2026 Aotearoa New Zealand Physical Oceanography Workshop NZ-POW



2-3 Sept 2026 ESNZ Greta Point Wellington <https://maps.app.goo.gl/ZsvGQZ7VTw9A76zP6>

Wednesday

Thursday

09:00	Craig Stevens	Welcome/karakia/history/objectives of POW		housekeeping	
	The heating ocean (chair Denise Fernandez)			Polar oceans (chair Andrew Pauling)	
09:15	Mitchell Chandler	Heatwaves been faking me out: Integrating HR-XBT and satellite observations to observe subsurface marine heatwaves		Alena Malyarenko	Who even know what's up with the Eastern Ross?!
09:30	Rafa Santana	Multi-Model Ensemble Forecasts of Sea Surface Temperatures and Marine Heatwaves for Aotearoa New Zealand		Xingyu Wang	An Idealised Numerical Framework for Ice-Shelf-Front Dynamics: Model Development and Preliminary Results
09:45	Julian Drake	Ocean Temperature Trends, Variability and Predictability Around NZ Over the Last Half Century		Kayo Takahashi	Meridional Energy Flux and Its Sensitivity to Southern Hemisphere Winds
10:00	Graham Rickard	What CMIP5 and CMIP6 Models Have Told Us About the Southwest Pacific Ocean Future Ocean States, and CMIP7 Prospects		Nick Mahon	Tidal Control of Dense Water Release from the Drygalski Trough
10:15	Plenary - Eleanor Chaos VUW - marine restoration and gaming (Chair Craig Stevens)			Monica Nelson (Lightning)	Circulation features observed in Vincennes Bay, East Antarctica, using novel SWOT techniques
10:30				Ken Hughes	Imaging thermocline microstructure with wave-pumped temperature sensors on moorings
	Morning tea + games (30 min)			Morning tea (25 min)	
	Shelf seas (chair Ali Della Penna)			Interdisciplinary work (chair Alena Malyarenko)	
11:00	Pete Russell	Of Taniwha and Underwater Tornosdos and "Alternative" Hairstyles.		Emma McGuigan	Zooplankton Vertical Distribution and Diel Vertical Migration in the Western South Pacific Subtropical Gyre
11:15	Hannah Melzer	Do Internal Waves Cool Reefs? Insights from Tawhiti Rahi / Poor Knights Islands		Qing Yang	Interannual variability of environmental conditions and phytoplankton dynamics in the Tīkapa Moana Te Moananui-ā-Toi / Hauraki Gulf: insights from a 14-year in situ time series
11:30	Erik Behrens	Waves from Space: An Atlas of Internal Waves Around New Zealand		Chao Han	Tropical cyclone impacts on ocean phytoplankton from global observations of satellite ocean color and BGC Argo profiles
11:45	Renske Koets	Generation and variability of internal waves near Rangitāhua		Alice Della Penna	Plankton Communities at Ocean Fronts: Insights from Open-Access Imaging Across Multiple Scales
12:00	Jack Beagley	Oceanographic Controls on Deep-water and Fjord Carbon Burial		Rob Smith	How do synoptic weather systems drive rapid mixing and re-stratification at a shelf break front? (5 min)
				Abby Nicholson	The role of mixed layer shoaling in generating marine heatwaves (5 min)
				Logan R. Chalmers	Resolving the shifting drivers of inter-annual stratification variability from a regional ocean model (5 min)
12:15	Lunch (45 min)			Lunch (45 min)	
13:00	Remy Zyngfogel	Operational coastal pollution response: the Wellington sewage spill event		Workshop discussion 2 - PO + "Tech development and wealth generation" - future directions?	
13:15	Craig Stevens	Cook Strait Te Moana o Raukawa - turbulence laboratory			
	Model development (chair Rafa Santana)				
13:30	Phyo Wai Thaw	The Response of the Ice Shelf–Ocean Boundary Layer to Shear and Thermal Forcing			
13:45	Andrew Pauling	P-SKRIPS Coupled Model Development: Integrating CICE Sea Ice into a Ross Sea Atmosphere–Ice–Ocean System			
14:00	Liv Cornelissen	How to force a small regional domain?			
14:15	Plenary: Deidre Cleland Data Science ESNZ - Accelerating Scientific Discovery with AI and Machine Learning				
14:45	Workshop discussion 1 - PO and data sci (60 min)			14:00 Closing & Karakia – Alena Malyarenko	
	Dinner @ 6 pm – The Botanist			TEAMS Link to follow	